

PMFBY Crop Insurance Analysis Dashboard

Analyzing crop insurance coverage, premium distribution, government subsidy patterns, and seasonal trends (2018–2022)



PROJECT OBJECTIVE:

To analyze PMFBY insurance coverage, subsidy allocation, farmer burden, crop distribution, and seasonal patterns across states and crops using Power BI, SQL and Python.

DATASET INFORMATION:

Dataset Source:

Kaggle

Years Covered:

2018–2022

Tools Used:

Python, SQL, Power BI

Records:

~20K PMFBY entries

KEY METRICS:

Total Sum Insured: Total crop value covered under insurance

Total Premium: Farmer + State + Central contribution

Govt Subsidy: State Share + GOI Share

Farmer Burden Percentage: Farmer contribution ÷ Total Premium

Premium to Coverage Ratio: Premium paid relative to insured value

KEY INSIGHTS FROM PMBFY(2018-2022):

Insurance coverage declined by 15% between 2018-2020.

Rabi season contributed more coverage than kharif.

Tamil Nadu, Uttar Pradesh, Maharashtra are top performing states.

Banana, Paddy and Wheat are among the most insured crops.

Agriculture Insurance Company dominates insurance coverage market share.



PMFBY Executive Overview (2018–2022)

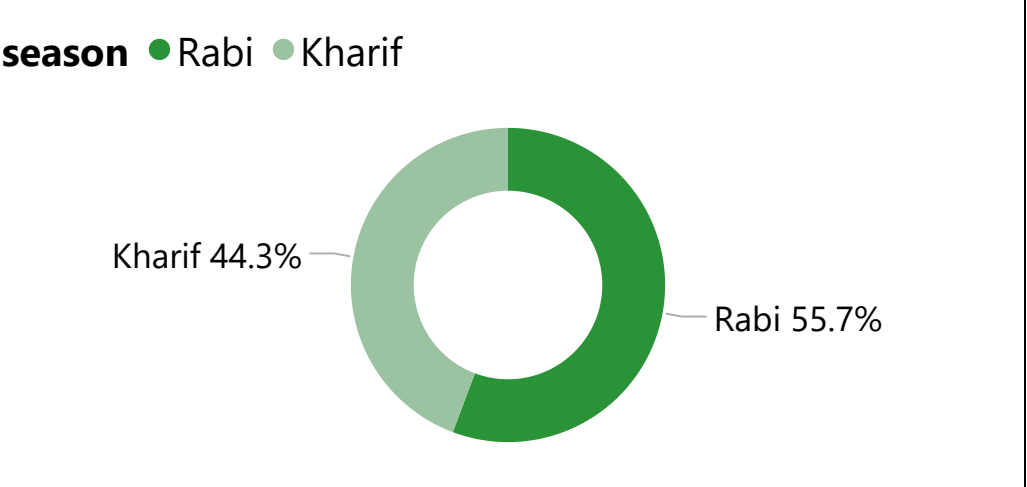


Total Sum Insured	Total Premium	Total Govt Subsidy	Average Farmer Burden %	Year
1.10bn	115.14M	86.66M	24.73%	All

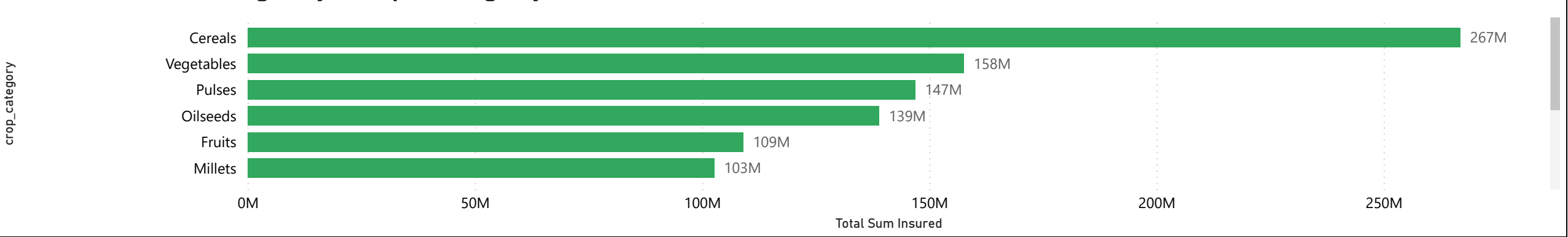
Top States by Insurance Coverage



Season-wise Insurance Distribution

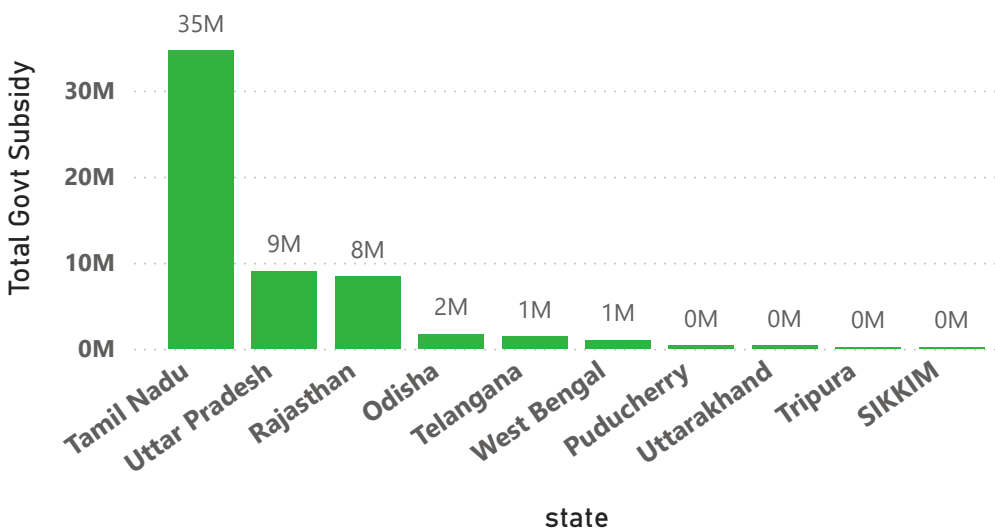


Insurance Coverage by Crop Category

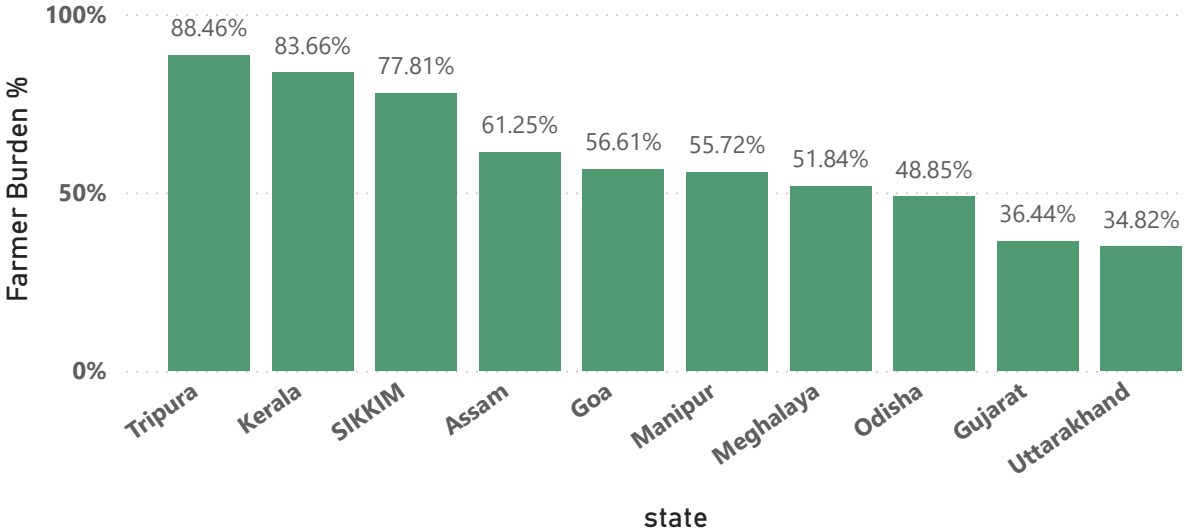


PMFBY State & District Analysis

Top 10 States by Govt Subsidy



Top 10 States by Farmer Burden %



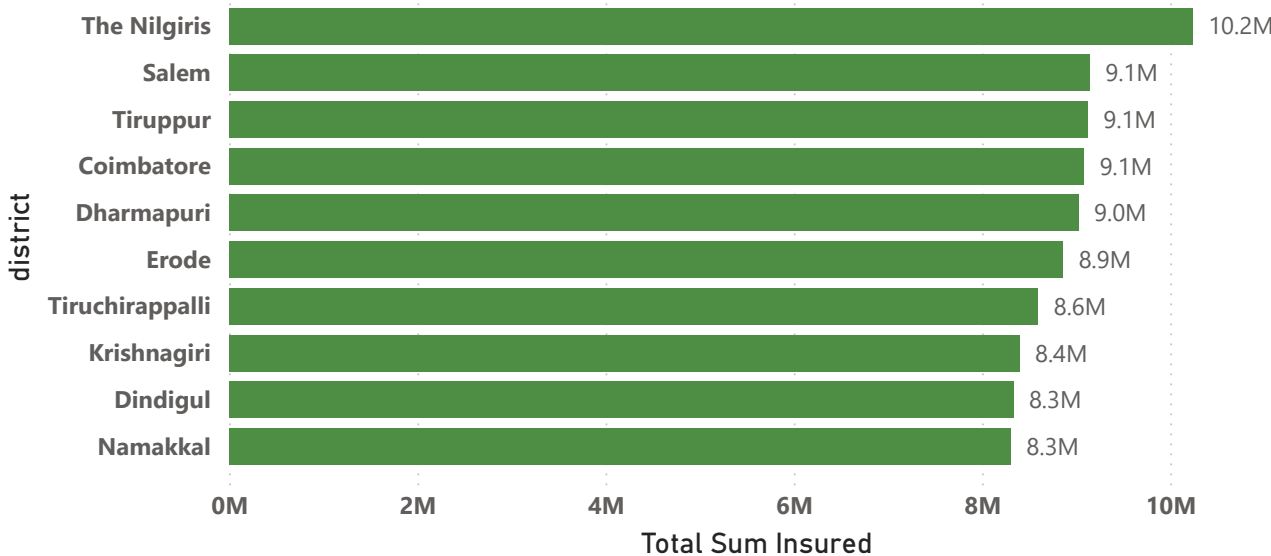
year

- 2018
- 2019
- 2020
- 2021
- 2022

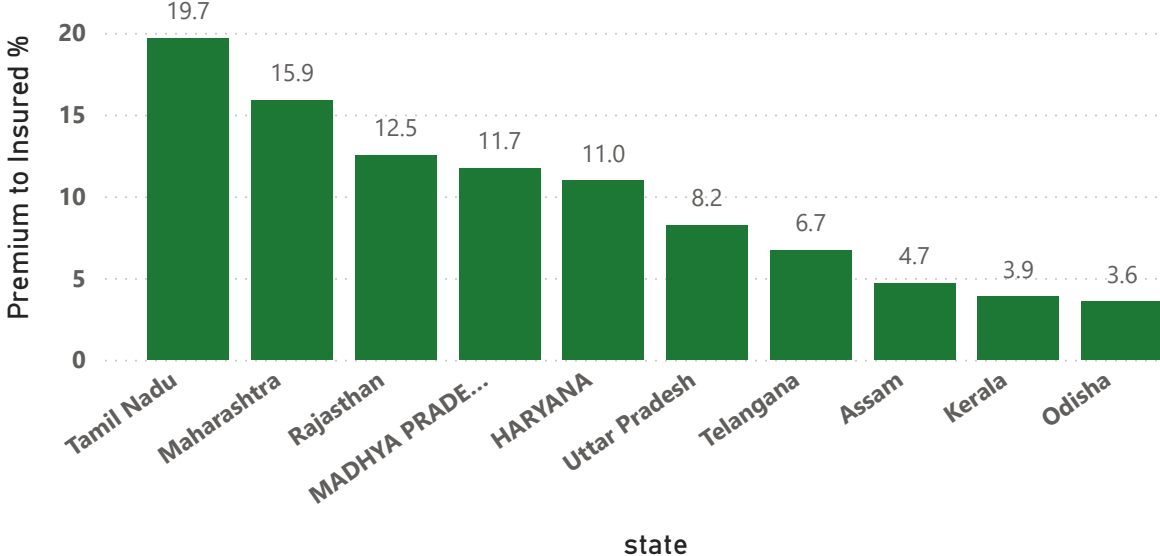
season

- Kharif
- Rabi

Top 10 districts by Insurance Coverage

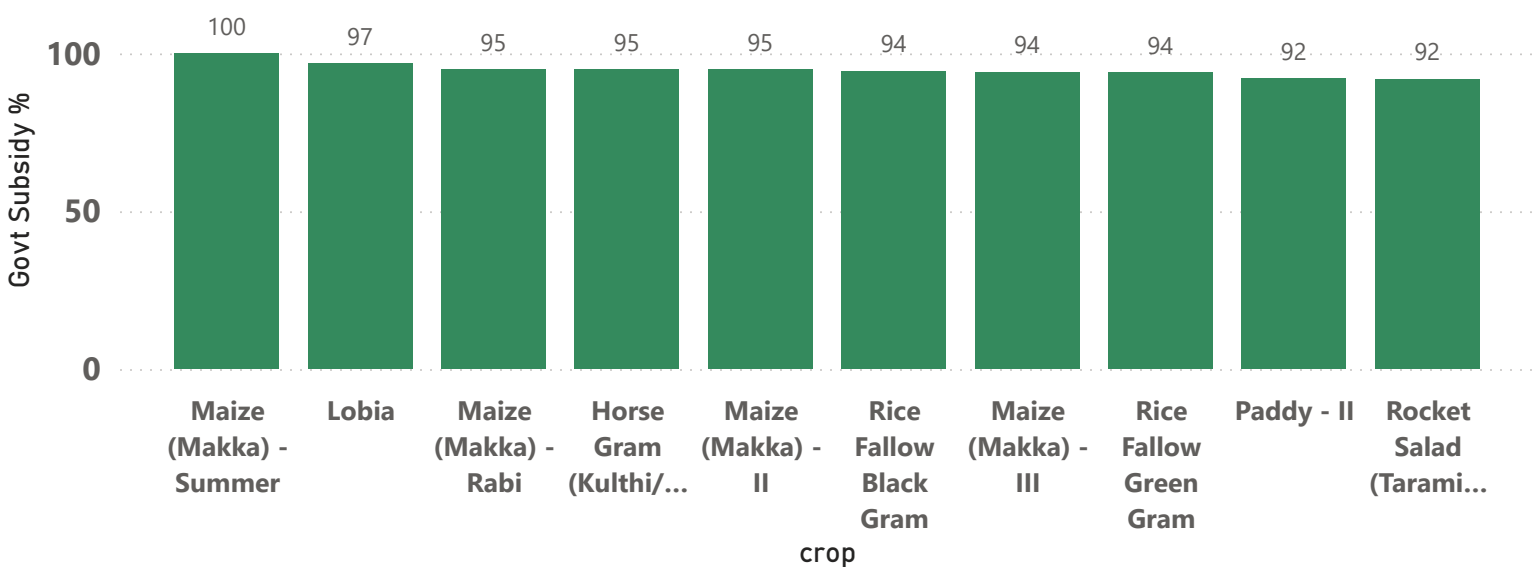


Top States by Premium to Coverage ratio



PMFBY Crop Analysis

Government Subsidy by Crop Category

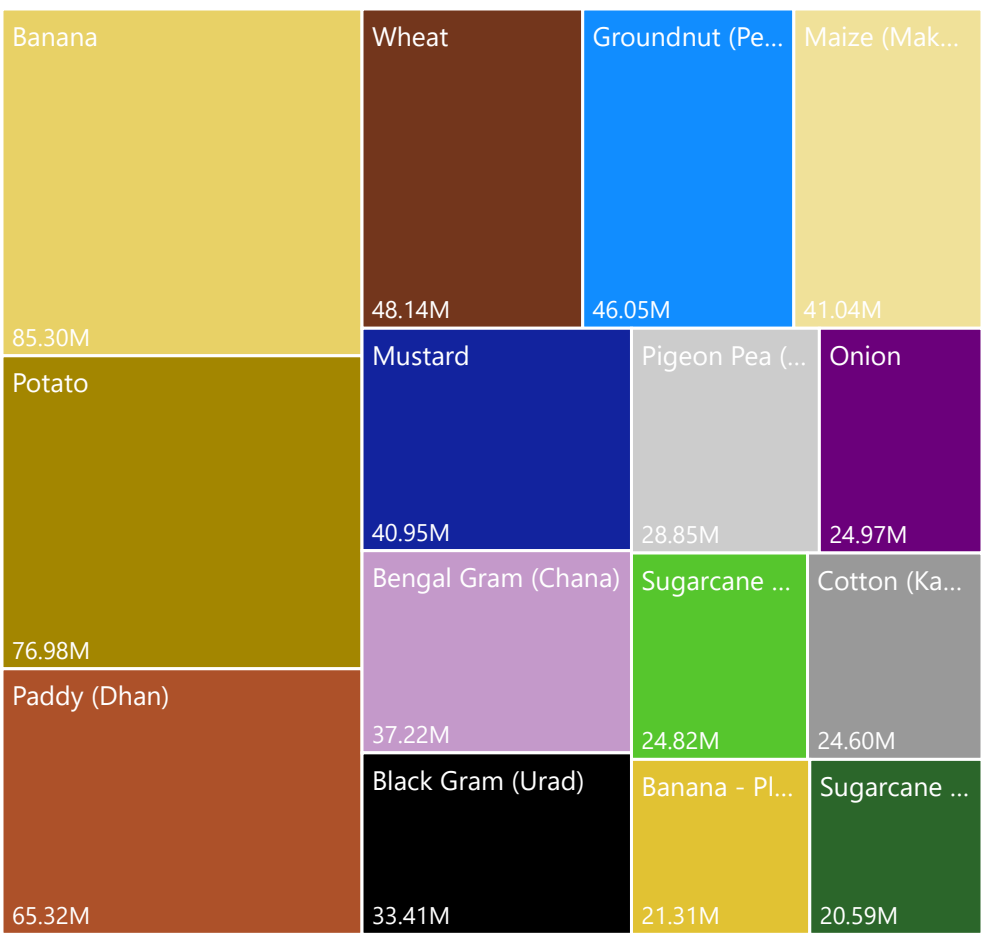


Season

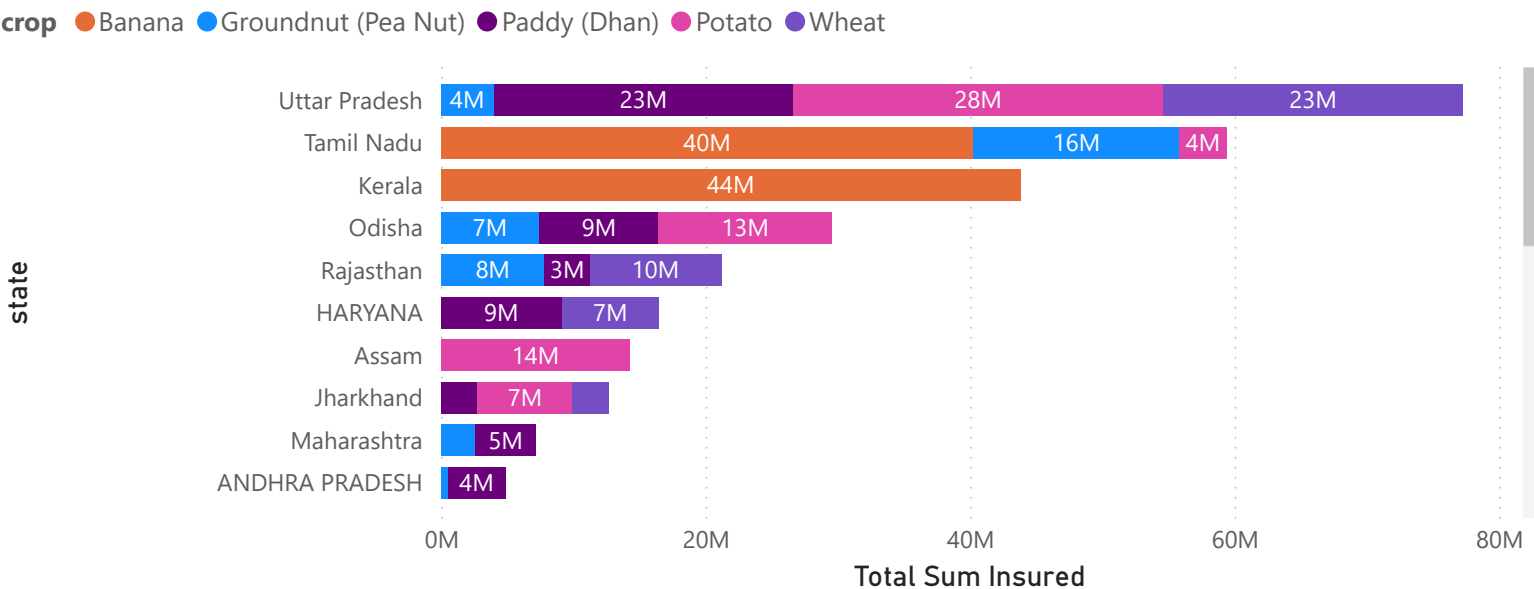
- Kharif
- Rabi

Insurance coverage was concentrated in a limited set of crops, with Banana, Potato and Paddy contributing major insured value. High subsidy support across crop categories indicates strong government participation in reducing farmer cost.

Crop Coverage Distribution

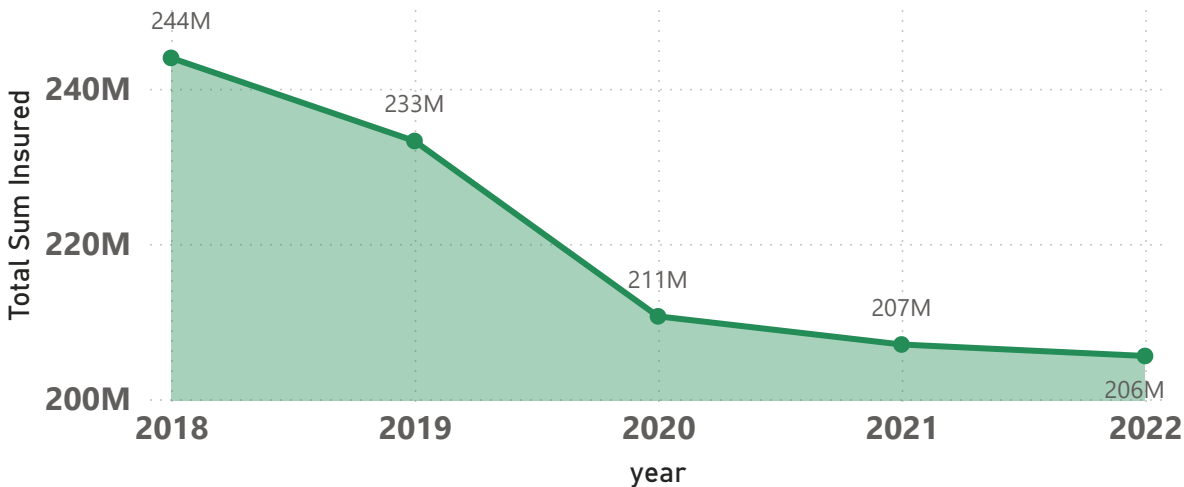


Top Sum Insured by state and crop



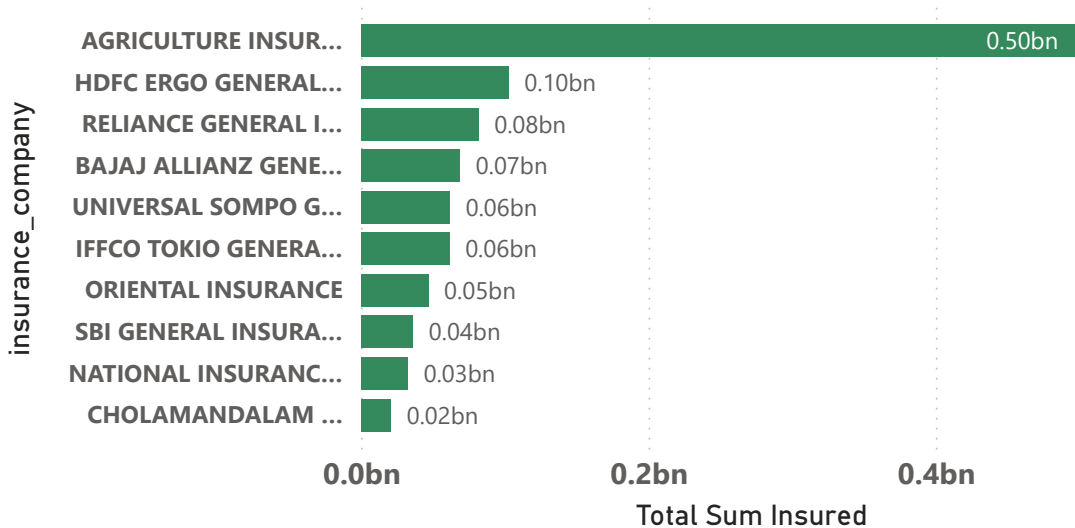
PMFBY Trends & Insurance Companies

Year-wise Insurance Coverage Trend



Coverage declined 15% from 2018–2022, indicating reduced scheme participation in later years

Top Insurance Companies by Insurance Coverage



season

☐ Kharif

☐ Rabi

year

☐ 2018

☐ 2019

☐ 2020

☐ 2021

Rabi season contributed higher insurance coverage than Kharif in leading states, with Tamil Nadu showing the strongest contribution.

Top Season–State Coverage Patterns

